

COMPUTER NETWORKS

Chapter- 1

EXERCISES

A) Multiple Choice Questions

- 1) Inter-connection of various devices is called.....
(a) chatting (b) networking (c) Internet **Ans: (b) networking**
- 2) Peripheral devices connected to the network are called
(a) servers (b) workstations (c) resources **Ans: (c) resources**
- 3) Which of the following is the most expensive network topology?
(a) Star topology (b) Mesh topology (c) Bus topology
Ans: (b) Mesh topology

B) Fill in the blanks.

- 1) A network can be of any _____ **Ans: size**
- 2) Your computer is a _____ on the network. **Ans: workstation**
- 3) A _____ is a device that connects dissimilar networks. **Ans: gateway**
- 4) A _____ is a collection of Local Area Network within a city. **Ans: MAN**
- 5) A _____ topology has a minimal line cost. **Ans: Bus topology**

C) Write the full forms of the following:

- 1) URL **Ans : URL Uniform Resource Locator**
- 2) PAN **Ans : PAN Personal Area Network**
- 3) LAN **Ans : LAN Local Area Network**
- 4) MAN **Ans : MAN Metropolitan Area Network**
- 5) WAN **Ans : WAN Wide Area Network**
- 6) HTTP **Ans: HTTP Hyper Text Transfer Protocol**
- 7) TCP/IP **Ans: TCP/IP Transmission Control Protocol / Internet Protocol**
- 8) Wi Fi **Ans: WIFI Wireless Fidelity**

D) Very Short Answer Questions

- 1) Name two types of wired media. Write which of them is more durable.
Answer- Two types of wired media are coaxial cable and Optical Fiber Cable. Optical Fiber Cable is more durable.
- 2) Who developed TCP/IP in 1978?
Answer- TCP/IP was developed by Vinton Gray Cerf and Robert Elliot Kahn in 1978.

E) Short Answer Questions

- 1) What is a node?
Ans- A node is a computer or any other device on a network that communicate with other devices.
- 2) What is topology?
Answer - The term network topology refers to the way in which the nodes (Computer or other devices that need to communicate) of a network are linked together. It determines the data paths, that may be used between any pair of nodes in the network.

3) Describe network in your own words.

Answer - A computer network is a collection of computers and peripherals devices connected by communication links.

4) Write three benefits of using a network

Answer - The three benefits of computer network are: -

- a) Easy sharing of files and information
- b) Sharing of hardware resources like printers
- c) Reduction in cost by sharing devices and software

F) Long Answer Questions

1. Differentiate between LAN and WAN.

Answer: - The difference between LAN and WAN are -

(a) LAN-A Local Area Network is the most common type of network. A LAN connects computer devices located close to one another, such as many computers in one building.

(b) WAN - A Wide Area Network connects LAN and MAN together. A Wide Area Network; may be located throughout a country or even around the world.

For example, Internet and ATM facility.

2. Write a note on the hardware parts required to establish a network.

Answer: -The hardware parts required to establish a network are:

- (a) MODEM - A MODEM (Modulator Demodulator) is an electronic device that enables a computer to transmit data over telephone lines. It is a device used to convert the digital signals into analog signals and vice versa.
- (b) HUB - A Hub is a hardware device that is used to connect several computers together. It provides a central connection point for cables from nodes, servers and peripherals. When a hub receives the data packets, it transmits those data packets to all the connected devices.
- (c) SWITCH - A Switch also connects computers on a network like a hub. But it sends the data packets only to the intended receiver.
- (d) NIC (Network Interface Card)- A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network and communicate with other devices.
- (e) Router- A Router is a networking device that connects different networks together and forwards data from one network to another.

3. Write the advantages and disadvantages of the following:

a) star topology

Ans - **Advantages of Star Topology**

- a) It has minimal line cost.
- b) If any of the local computers fails, the remaining portion of the network remains unaffected.

Disadvantages of Star Topology

- a) The system entirely depends on the central switch. If it fails, the entire network goes down.

b) Ring Topology

Ans: - **Advantages of Ring Topology**

- a) It works well where there is no central site computer system.
- b) It is more reliable than the star topology.

Disadvantages of Ring Topology

- a) It requires more complicated control software.
- b) Failure of one node results in the failure of the entire network.

c) Bus Topology

Ans: - Advantages of Bus Topology

- a) It is quite easy to set up.
- b) Failure of one node does not affect the rest of the network.

Disadvantages of Bus Topology

- a) It offers limited flexibility for change.
- b) A signal on the bus must be strong enough to reach the receiver.

d) Mesh Topology

Ans: - Advantages of Mesh Topology

- a) Failure of one node does not affect the rest of the network.
- b) Communication is very fast between any two nodes.
- c) Changes in the network are done easily without interrupting other nodes.

Disadvantages of Mesh Topology

- a) It is the most expensive network as there are many redundant connections.
- b) Maintenance of this network is very difficult.

4. What do you mean by wireless communication? Explain any one popular wireless technology.

Answer - Wireless means there is no physical path for the signals (data) to move, wireless communication is one of the biggest contributions to humankind. It involves sending and receiving information over a distance without using wires or cables. It is also known as unguided Media. Some of the wireless media are: Bluetooth, Infrared and WIFI (Wireless Fidelity).

G) Computer In Everyday Life (Application based Questions)

1) The principal of a school wants to connect all the computers and devices within the school building. Which network should he/she use to do the same?

Ans: The principal may use Campus Area Network to connect all devices within a school.

2) Navleena is interested in transferring a few songs from her mobile phone to her father's mobile phone. Suggest to her the simplest wireless option for doing the same.

Ans: Navleena can use bluetooth to transfer songs from her mobile to father's mobile.

ADDITIONAL QUESTIONS WITH ANSWERS

Short Answer Questions with Answers

1) What is a network?

Answer- A network is a group of devices connected together to share data and resources.

2) Define computer network.

Answer- A computer network is a collection of computers and peripheral devices connected through communication links.

3) What is a file server?

Answer- A file server is a computer that stores and distributes files and programs to users on a network.

4) What is a workstation?

Answer- A workstation is a computer connected to a network.

5) What is protocol?

Answer- Protocol is a set of rules used for communication between network devices.

- 6) What is networking media?
Answer- Networking media is the communication path through which data is transferred.
- 7) What is bandwidth?
Answer- Bandwidth is the amount of data transferred in a fixed amount of time.
- 8) What is a website?
Answer- A website is a collection of related webpages.
- 9) What is a webpage?
Answer- A webpage is a single page of a website.
- 10) What is a hyperlink?
Answer- A hyperlink is a clickable text or image that opens another webpage.
- 11) What is URL?
Answer- URL stands for Uniform Resource Locator. It is the address of a webpage.
- 12) What is an IP address?
Answer- An IP address is a unique numeric address assigned to a computer on a network.
- 13) Who designed and published the File Transfer Protocol (FTP) in 1971?
Answer: Abhay Bhushan designed and published FTP in 1971.
- 14) Which institutions did Abhay Bhushan study at?
Answer: He studied at Indian Institute of Technology Kanpur and Massachusetts Institute of Technology.
- 15) What does FTP stand for?
Answer: FTP stands for File Transfer Protocol.
- 16) In which year was FTP designed and published?
Answer: FTP was designed and published in 1971.
- 17) Who developed TCP/IP?
Answer: Vinton Gray Cerf and Robert Elliot Kahn developed TCP/IP.
- 18) In which year was TCP/IP developed?
Answer: TCP/IP was developed in 1978.
- 19) Which protocol was developed earlier, FTP or TCP/IP?
Answer: FTP was developed earlier.
- 20) What is the contribution of Abhay Bhushan in computer networking?
Answer: He designed and published the File Transfer Protocol (FTP).

Long Answer Questions with Answers

1) Explain computer network and its benefits.

Answer- A computer network is a group of computers and devices connected together to share information and resources.

Benefits of Computer Network:

- a) **Ease of Access** – Users can easily access files and information from any computer on the network.
- b) **Work from Home** – Employees can connect to the office network and work from home.
- c) **Sharing Resources** – Devices like printers and hard drives can be shared among users.
- d) **Sharing Information** – Users can quickly exchange files, documents, and data.
- e) **Sharing Programs** – Software can be installed on one central computer and used by many users.
- f) **Security** – Usernames and passwords protect network information from unauthorized access.
- g) **Reduced Cost** – Sharing hardware and software lowers overall expenses.
- h) **Increased Productivity** – Faster communication and resource sharing improve work efficiency.

2) Explain the components of data communication.

Answer- The components of data communication are:

- a) **Message**- The information being sent such as text, audio, image or video.
- b) **Sender**- The device that sends the message.

- c) **Receiver-** The device that receives the message.
- d) **Communication Medium-** The path through which data travels such as cables or wireless signals.
- e) **Protocol-** The rules used for communication between sender and receiver.

3) Describe wired and wireless media with examples.

Answer- **Wired Media-** Wired media uses cables for data transfer.

Examples: Twisted Pair Cable , Coaxial Cable, Optical Fibre Cable

Wireless Media- Wireless media transfers data without wires.

Examples: Bluetooth, Infrared, WiFi

4) Explain different types of computer networks.

Answer-

- a) **PAN (Personal Area Network)-** Used for connecting devices within a few metres.
- b) **LAN (Local Area Network)-** Connects computers in a building or office.
- c) **CAN (Campus Area Network)-** Connects multiple LANs within a campus.
- d) **MAN (Metropolitan Area Network)-** Connects networks within a city.
- e) **WAN (Wide Area Network)-** Connects networks over very large distances. Internet is an example.

5) What is network topology? Explain its types.

Answer- Network topology refers to the arrangement of computers and devices in a network.

Types of Topology:

- 1. Star Topology
- 2. Ring Topology
- 3. Bus Topology
- 4. Mesh Topology

Each topology has different advantages and disadvantages.

6) Explain IP address and DNS.

Answer- **IP Address-** An IP address is a unique number assigned to each computer connected to a network.

Example: 192.168.1.1

DNS- DNS converts domain names into IP addresses.

Example:

www.google.com → IP address

7) Explain Internet Protocol, TCP and HTTP.

Answer-

- a) **Internet Protocol (IP)-** Used for sending data between computers.
- b) **Transmission Control Protocol (TCP)-** Arranges data packets in correct order.
- c) **Hypertext Transfer Protocol (HTTP)-** Used for transferring webpages on the Internet.

TCP/IP: TCP and IP together are called TCP/IP, which is the main communication protocol of the Internet.

MCQs with Answers – Computer Networks

1. A computer connected to a network is called:

- (a) Server (b) Workstation (c) Protocol (d) Modem

Answer: (b) Workstation

2. Which is the largest computer network?

- (a) PAN (b) LAN (c) Internet (d) MAN

Answer: (c) Internet

3. Which topology uses a central switch?

- (a) Ring (b) Bus (c) Star (d) Mesh

Answer: (c) Star

4. WiFi is an example of:

- (a) Wired media (b) Wireless media (c) Cable (d) Optical fibre

Answer: (b) Wireless media

5. DNS converts:

- (a) Passwords (b) Files (c) Domain names into IP addresses (d) Images

Answer: (c) Domain names into IP addresses

6. HTTP is used for:

- (a) Webpage transfer (b) Printing (c) Scanning (d) Gaming

Answer: (a) Webpage transfer

7. Which cable provides high-speed data transfer?

- (a) Twisted pair cable (b) Optical fibre cable (c) USB cable (d) Telephone wire

Answer: (b) Optical fibre cable

8. Which network covers a city?

- (a) PAN (b) LAN (c) MAN (d) CAN

Answer: (c) MAN

9. A private network inside an organisation is called:

- (a) WAN (b) Intranet (c) Internet (d) DNS

Answer: (b) Intranet

10. The rules for communication are called:

- (a) Media (b) Protocol (c) Browser (d) Website

Answer: (b) Protocol